Results (Proof, left to the reader)

(1) Two tangents can be drawn to (i) a parabola (ii) an ellipse and (iii) a hyperbola, from any external point on the plane.

(2) Four normals can be drawn to (i) an ellipse and (ii) a hyperbola from any external point on the plane.

(3) The locus of the point of intersection of perpendicular tangents to

(i) the parabola $y^2 = 4ax$ is $x = -a$ (the directrix).

(ii) the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 + b^2$ (called the director circle of ellipse).

(iii) the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 - b^2$ (called director circle of hyperbola).

Example 5.29

Find the equations of tangent and normal to the parabola $x^2 + 6x + 4y + 5 = 0$ at $(1, -3)$.

Solution

Equation of parabola is $x^2 + 6x + 4y + 5 = 0$.

$$x^2 + 6x + 9 - 9 + 4y + 5 = 0$$

$$(x+3)^2 = -4(y-1) \quad \ldots (1)$$

$$\text{Let } X = x+3, Y = y-1$$

Equation (1) takes the standard form

$$X^2 = -4Y$$

Equation of tangent is $XX_1 = -2(Y + Y_1)$

At $(1, -3)$ $X_1 = 1+3 = 4; Y_1 = -3-1 = -4$

Therefore, the equation of tangent at $(1, -3)$ is

$$(x+3)4 = -2(y-1-4)$$

$$2x + 6 = -y + 5 .$$

$$2x + y + 1 = 0 .$$

Slope of tangent at $(1, -3)$ is -2, so slope of normal at $(1, -3)$ is $\frac{1}{2}$

Therefore, the equation of normal at $(1, -3)$ is given by

$$y + 3 = \frac{1}{2}(x-1)$$

$$2y + 6 = x - 1$$

$$x - 2y - 7 = 0 .$$

Example 5.30

Find the equations of tangent and normal to the ellipse $x^2 + 4y^2 = 32$ when $\theta = \frac{\pi}{4}$.

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